

Linearity and the difference form of the canonical coefficients

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Abstract

Completion of Astra #10's R6. On the envelope class the canonical coefficients are linear: $A_\alpha(c-d) = A_\alpha(c) - A_\alpha(d)$ and $B_\alpha(c-d) = B_\alpha(c) - B_\alpha(d)$ (`canonA_sub`, `canonB_sub`), proved through the collision formula and the moment series of unit 126 (the finite-part functionals are linear on absolutely convergent rows; the envelope class is closed under differences with envelope $H + H'$). With the weighted-norm bounds of unit 140 this gives the difference form Astra asked for: $|A_\alpha(c) - A_\alpha(d)| \leq w_C(\alpha) \|c - d\|_{\alpha,0,r}$ and $|B_\alpha(c) - B_\alpha(d)| \leq (w_U + w_V)(\alpha) \|c - d\|_{\alpha,0,r} + w_C(\alpha) \|c - d\|_{\alpha,1,r}$ (`abs_canonA_sub_le`, `abs_canonB_sub_le`). Zero sorry/axiom.

1 The things formalised

```
theorem logMoment_sub ( : ) ( : ) (c c : → )
  (h : IntegrableOn ...) (h : ...) :
  logMoment      (fun s => c s - c s) = logMoment      c - logMoment      c
theorem sub_envelope (c d : × → → ) ( : ) (h : 0 < ) (H H' : → )
  (hc : ij s, |c ij s| * ^ (ij.1 + ij.2) H s)
  (hd : ij s, |d ij s| * ^ (ij.1 + ij.2) H' s) (ij : × ) (s : ) :
  |c ij s - d ij s| * ^ (ij.1 + ij.2) H s + H' s
theorem canonA_sub ... ( : ) (h : 0 < ) :
  canonA h h k k (fun i j s => c (i, j) s - d (i, j) s)
  = canonA h h k k (fun i j s => c (i, j) s)
  - canonA h h k k (fun i j s => d (i, j) s)
theorem canonB_sub ... ( : ) (h : 0 < ) :
  canonB b h h k k (anaFaceU (fun ij s => c ij s - d ij s) b)
  (anaFaceV (fun ij s => c ij s - d ij s) b)
  (fun i j s => c (i, j) s - d (i, j) s)
  = canonB ... (c) - canonB ... (d)
theorem abs_canonA_sub_le ... ( : ) (h : 0 < ) :
  |canonA ... (c) - canonA ... (d)|
  cWeight r h h k k * coeffNorm 0 r (fun ij s => c ij s - d ij s)
theorem abs_canonB_sub_le ... ( : ) (h : 0 < ) :
  |canonB ... (c) - canonB ... (d)|
  (uWeight b r h h k k + vWeight b r h h k k )
  * coeffNorm 0 r (fun ij s => c ij s - d ij s)
  + cWeight r h h k k * coeffNorm 1 r (fun ij s => c ij s - d ij s)
```

2 Mathematics

Linearity is immediate for the collision coefficient and for the log moment (Bochner integrals of integrable integrands). For the face moments the direct definition through the finite part of the

face function is awkward; instead the moment series $k_1^{-1} \sum_j q_j M[\alpha; \ell; c_{ij}]$ of unit 126, valid for c , d and $c - d$ (all in the envelope class), reduces linearity to $M[c_{ij} - d_{ij}] = M[c_{ij}] - M[d_{ij}]$ and the linearity of absolutely convergent sums. The difference bound is then the boundedness theorem of unit 140 applied to $c - d$.

3 Paper-facing meaning

The coefficient functionals $c \mapsto (A_\alpha(c), B_\alpha(c))$ are bounded linear functionals of the amplitude coefficient array in the weighted moment norm, with explicit constants: the deterministic half of the continuity argument that turns $\xi_n \Rightarrow G$ into convergence of the Taylor-tree coefficients. The remaining nonlinear link is the amplitude map $(x, y) \mapsto c = e^{\beta s x_{00}}(y * E_{x-x_{00}}(\beta s))$, whose local Lipschitz estimate in the coefficient Banach-algebra norm is Astra #10's rank 5.

4 Lean notes

- `tsum_sub` is protected: use `hf.tsum_sub hg` (dot notation).
- `le_or_lt` is gone (`le_or_gt`); here the case split was unnecessary once $\rho > 0$ was made a hypothesis.
- Proving linearity through a series identity valid on a class closed under differences avoids any linearity lemma for the finite-part functional itself.